# **SuRe**: Summarizing Retrievals using Answer Candidates for Open-domain QA of LLMs

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🤷: "I want you to provide **more accurate answer** and **rationale to support this**, but with a **simple method**. Is it possible?"

🤖: "**SuRe**, if you ask me through **SuRe**!"

Paper

Code

## 0 3 Sentence Summary of This Work

❒ **Why?** Easy-to-use, applicable to any LLMs, and provides more accurate answer and supporting rationale

❒ **How?** 3 steps of prompting (*i.e.*, **no training**!):
(1) Generating answer candidates, (2) Conditional summarization of retrieval, and (3) Selection with verification

❒ **What?** 4.6%/4.0% avg. improvement of EM/F1 across various QA benchmarks, LLMs, and retrieval methods. Also, SuRe's summarization serves as strong rationale for answer

## 1 Introduction

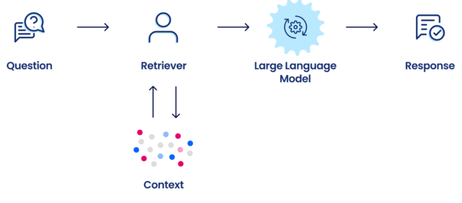
❒ Challenges with Large Language Models (LLMs)

➢ Non-factual but seemingly plausible generation (*i.e.*, **Hallucination**[1])

➢ Difficulty in integrating **up-to-date knowledge**

❒ Retrieval Augmentation with LLMs

➢ Retrieving and augmenting relevant knowledge → **promising solution**




*Concept Figure of Retrieval-augmented Generation*[2]

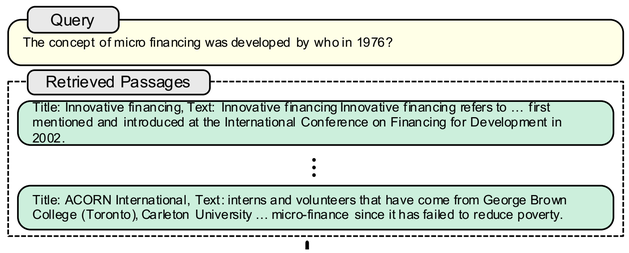
❒ Challenges for Retrieval Augmentation with LLMs

➢ Large scale/black-box nature → **Separate training**[2] **is infeasible**

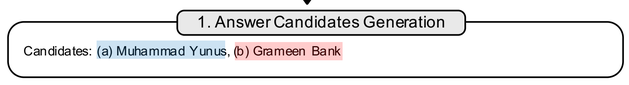
➢ Simple prompting by appending → **Limited exploitation of retrieval**[3]

## 2 Main Method: SuRe

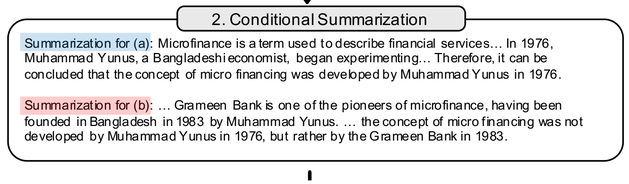
❒ **Step 0**. Preparing question and retrieval documents



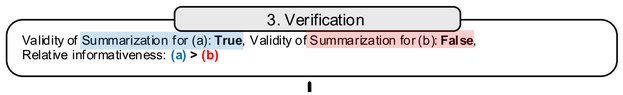

❒ **Step 1**. Generating multiple answer candidates




❒ **Step 2**. Generating summarization to support each candidate




❒ **Step 3**. Self-verification and ranking summarizations




❒ **Result**. Selecting candidate with most plausible summarization




## 3 Experimental Results

❒ Comparison with Baselines on ChatGPT and BM25

➢ Details: 0-shot, 10 retrieved passages, (EM / F1)

| Methods / Datasets | NQ | WebQ | 2Wiki | HotpotQA | Average |
|---|---|---|---|---|---|
| No retrieval | 27.6 / 39.0 | 25.0 / **38.8** | 21.4 / 24.8 | 22.2 / 31.9 | 24.1 / 33.6 |
| Base | 28.4 / 38.8 | 19.6 / 32.5 | 27.4 / 32.8 | 30.8 / 40.3 | 26.6 / 36.1 |
| Rerank | 24.8 / 33.9 | 18.8 / 30.6 | 23.0 / 28.4 | 27.8 / 37.4 | 23.6 / 32.6 |
| RePlug | 26.0 / 35.3 | 18.8 / 31.5 | 23.6 / 28.5 | 28.0 / 37.9 | 24.1 / 33.3 |
| Selection-inference | 24.3 / 32.8 | 17.3 / 28.6 | 22.6 / 29.5 | 30.8 / 39.6 | 23.8 / 32.6 |
| Chain-of-thoughts | 22.3 / 31.4 | 15.2 / 27.8 | 19.6 / 22.5 | 25.6 / 31.8 | 20.7 / 28.4 |
| Self-verification | 25.2 / 35.4 | 16.1 / 28.5 | 23.2 / 30.5 | 31.6 / 41.8 | 24.0 / 34.1 |
| SuRe (Ours) | **33.5 / 42.3** | **25.1** / 36.6 | **32.8 / 38.1** | **33.2 / 43.4** | **31.2 / 40.1** |

❒ Compatibility with Various LLMs and Retrievals

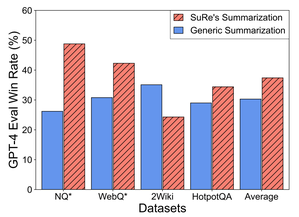
➢ Exact Match (EM)

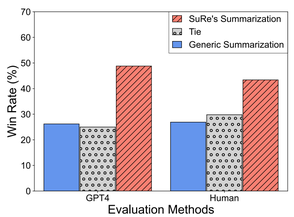
| | ChatGPT | | | | | | GPT-4 | | LLaMA2-chat | |
|---|---|---|---|---|---|---|---|---|---|---|
| Datasets | BM25 | + SuRe | DPR | + SuRe | Contriever | + SuRe | BM25 | + SuRe | BM25 | + SuRe |
| NQ | 28.4 | **33.5** | 36.1 | **41.0** | 35.8 | **40.4** | 30.2 | **32.4** | 18.6* | **30.4*** |
| WebQ | 19.6 | **25.1** | 23.2 | **27.3** | 22.5 | **28.7** | 21.5 | **21.7** | 16.0* | **24.0*** |
| 2Wiki | 27.4 | **32.8** | 19.2 | **21.4** | 27.2 | **32.6** | 34.8 | **38.2** | 20.2 | **27.8** |
| HotpotQA | 30.8 | **33.2** | 25.6 | **27.4** | 32.2 | **33.6** | 34.8 | **40.6** | 24.0 | **28.0** |
| Average | 26.6 | **31.2** | 26.0 | **29.3** | 29.4 | **33.8** | 30.3 | **33.2** | 19.7 | **27.6** |

❒ Advantages of SuRe as Explicit Rationale for Answer

➢ We compare which summarization is better to support same answer

➢ Both GPT-4[4] and Human prefer SuRe's summarization! (7%,16%)




❒ References

[1] Welleck et al., Neural Text Generation with Unlikelihood Training., ICLR 2020
[2] Guu et al., REALM: Retrieval-Augmented Language Model Pre-Training., ICML 2020
[3] Liu et al., Lost in the Middle: How Language Models Use Long Contexts., arXiv:2307
[4] Zheng et al., Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena., NeurIPS 2023 Dataset & Benchmark